



Samsung's unbreakable panel

Samsung has built an unbreakable smartphone display, which has been certified by the US DoD.
<https://dgit.in/Aug18-21-1>

graphics tech at Hot Chips (an enterprise symposium) then there has to be a proper public unveil within the same time window. Gamescom is the only major event that fits the bill.

NVIDIA TURING GTX 1180 / 1170 SPECIFICATIONS

Usually, we see NVIDIA come out with one large GPU which embodies the peak of a new architecture. With Pascal, this was the GP100 GPU. A slightly scaled down version was present on the Pascal-based NVIDIA Tesla compute card. We then had the GP102, GP104 and GP106 GPU variants which were all used for different graphics cards, along with sub-variants. E.g the GP102 has three sub-variants - the GP102-350, GP102-400 and GP102-450 with variations in terms of core count, memory and clock speeds. The GP102-350 lies at the heart of the GTX 1080 Ti, the GP102-400's in the NVIDIA TITAN X and the GP102-450's in the TITAN Xp.

All of them are derived from the GP100. So now, let's see if we can derive the GT104/GV104 from the NVIDIA Volta-based GV100 GPU that was unveiled last year in. You can find our analysis here - <https://dgit.in/NVIDIA11xx>.

NVIDIA TURING GTX 1180 / 1170 / 1160 PERFORMANCE

The only leak regarding performance of the new GPUs has been an image posted on a Polish hardware forum, FrazPC. It shows a GTX 1170 along with an Intel Core i5-8600K scoring 29,752 in 3DMark. The screenshot mentions that the result has been validated online i.e. the result should be recorded in 3DMark's online database, although no records are traceable now. The card seems to have 16 GB of buffer memory and a clock speed of 2512 MHz, which is a huge improvement over Pascal. The GTX 1070 has a core clock of 1506 MHz. So we see exactly double that here making this quite unbelievable. A score of 29,752 puts it ahead of the GeForce GTX 1080 Ti. Yes, if this is in fact proper, then the GTX 1170 beats every gaming card released till now.

The screenshot could be fake, according to multiple publications. While NVIDIA could be behind the absence of the scores from the online database, the huge difference in the core clocks is a bit difficult to believe. The highest clocked GTX 1080 Ti is the COLORFUL GeForce GTX 1080 Ti 11GB iGame KUDAN with a boost clock of 1784 MHz. A 728 MHz increment is supernormal. Despite all that, if we are to look at the last two generations, then the GTX 1070 did beat the GTX 980 Ti in a lot of benchmarks. And that makes this all the more believable.

NVIDIA GTX 1060 VS GTX 980 VS GTX 960

Also, if the GTX 1170 beat the GTX 1080 Ti, then we can see the GTX 1160 approximately equaling the GTX 1080 in performance the way the GTX 1060 beat the GTX 980. What remains is the GTX 1180 and that would end up being about 30-40% better than the GTX 1080 if we are to follow the same logic, which is what NVIDIA's GPUs have followed since the introduction of Maxwell.

NVIDIA TURING GTX 1180 / 1170 / 1160 PRICE

Given that NVIDIA and all board partners over-estimated the GPU mining craze and are now left with a massive surplus of GPUs, they have incurred a loss. According to a report by SemiAccurate, NVIDIA's been aggressively buying GDDR5 memory from vendors so that they can build graphics cards out of the excess GPU inventory and sell them before the GTX 11xx launch. Then there's the simple fact that the Radeon Vega 64 was unable to beat NVIDIA's flagship offerings. So NVIDIA has no reason to keep the price competitive.

These three reasons point to a slightly inflated price tag for the GTX 1180. As per WccfTech, they've got an inkling from a source that the Founders Edition of the GTX 1180 will be priced at \$699. That's a whole \$100 greater than the GTX 1080's launch price. Since there is no evidence to back this up, we're going to take this with a pinch of salt. Also, we're yet to hear anything regarding the pricing of the GTX 1170 and the 1160. 



DATA STORAGE INSIGHTS

New Approaches for A New Data Age

According to a report by IDC and sponsored by Seagate, data volume will grow to an astounding 163 zettabytes (ZB) by 2025, a ten-fold increase over what was created in 2016. Given below are five key disruptive data trends reshaping the global datasphere, as identified by the report.

From Life-Enhancing to Life-Critical

Data has become essential to the optimal operation of our society and our lives. Were the data flow to stop, not only would our business operations cease, but also the smooth operation of our daily lives. IDC's report estimates that by 2025, 20% of all data will be critical to our lives.

Embedded Systems and the Internet of Things (IoT)

Big Data and metadata continue to grow. The average rate per capita of data-driven interactions per day is expected to increase twentyfold in the next ten years as our homes, workplaces, appliances, vehicles and wearables become data-enabled.

True Mobile and Real-Time data

By 2025, connected users are expected to number 75% of the world's population, including previously unconnected groups like young children, the elderly, and people in emerging markets. These improvements in connectivity require parallel leaps in real-time and mobile data access.

Machine Learning Changes the Landscape

Thanks to staggering growth in analysable data, cognitive systems will become proactive drivers of action. Technologies such as machine learning, natural language processing and artificial intelligence hold the potential to make data analysis more frequent, flexible and important to our lives.

Security as a Critical Foundation

As the global datasphere explodes in size, so does the gap between the amount of data that should be secured and the amount that actually is. Data requiring security includes corporate financial data, personally identifiable information (PII), and medical records.

As the world's leading data storage brand, Seagate is at the forefront in developing radically innovative technologies to help customers manage the gigantic data volumes. Watch this space to learn more about innovations that will power efficient storage in the New Data Age.



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